

ENGR 102 Summer Bridge

Day 15 Student Quiz

Append, Concatenation, Slicing, and Strings as Sequences

20 points • approximately 20-25 minutes

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Boolean-operator convention. Python operators appear as `and`, `or`, and `not`. Their lowercase spelling is required in code.

Show intermediate state for trace credit. Program credit requires one coherent solution. Unless a prompt says otherwise, give exact Python 3 output.

Question 1. Trace sequence state and direct slicing.

4 points

```
text = "bridge"
pieces = []
for index in range(0, len(text), 2):
    pieces.append(text[index])
middle = text[1:5]
result = pieces + [middle]
print(pieces)
print(middle)
print(result)
```

Give the index ledger and every exact output line.

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Question 2. Repair sequence concatenation.

3 points

```
items = ["x"]
items = items + "y"
print(items)
```

Explain the failure and give both a concatenation repair and an append repair.

Question 3. Program construction

13 points

Write one complete Python program that satisfies every requirement.

- Read word as text and print invalid if its length is less than 3.
- Otherwise append characters at odd indexes to odd_characters.
- Count lowercase t characters.
- Build frame from the first one-character slice plus the last one-character slice.
- Print odd_characters, t_count, and frame. Include an iteration test with expected output.

[illegible]

Test evidence / expected result